PingSheng Kao

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SUMMARY

Applied AI and ML Data Scientist with 5+ years of experience translating user and business needs into AI evaluation systems, quality metrics, internal production tools, and evidence-based development decisions. Builds cross-functional evaluation workflows spanning researchers, engineers, Data Operations, and leadership.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Developed an auditable, expert-aligned LLM-as-judge evaluator deployed in an internal production workflow for performance reporting and failure discovery, reaching 90% alignment with experts on a 40-example gold set.
- Translated user-research needs into engineering requirements and labeling instructions for Data Operations, coordinating LLM-as-judge evaluation with human-in-the-loop validation.
- Built simulated-user, agent-as-judge, targeted-scenario, and behavioral-metric workflows for a multimodal conversational agent; systematic failure patterns surfaced by evaluation changed subsequent development direction.
- Owned designated evaluators and shipped versioned updates; scaled LLM- and agent-as-judge runs on an existing internal batch platform, resolving throughput and rate-limit issues and surfacing results in dashboards.
- Produced leadership-facing model-release reports combining quality metrics, regression attribution, embedding-cluster diagnosis, and online validation to make systematic risks and tradeoffs decision-ready.
- Wrote SQL in an evaluation pipeline querying AWS Athena to calculate model-quality metrics and export results to dashboards.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments to ECCV-accepted research on grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming: Python, SQL, C++, JavaScript
Applied AI: LLM-as-Judge, Agent-as-Judge, User Simulation, AI Prototype Evaluation, DSPy
Evaluation: Quality Metrics, Experiment Design, Human Evaluation, Failure Analysis, Responsible AI
Data & Workflows: AWS Athena, Embeddings, Clustering, OOD Analysis, Tableau, Evaluator Versioning